

Session 9

Integration:

The Game Console



Make a game on a one-row screen.

Part	Topic	Time
01	A strip is a one-row screen	6 min
02	Drawing the game state	8 min
03	Practice: Strip Catcher — move, catch, score	12 min
04	Make it a console: restart, speed, and what's coming	4 min

Builds on joystick zones and edges from Session 5, the for loop from Lesson 4, and states from M2 Session 4.

Five words you will hear today.

Word	Plain meaning
Pixel	One dot of light on the screen — on an LED strip, one LED.
Game state	Every number that describes the game right now: where you are, where the fruit is, the score.
Draw	Turn the game state into what the player sees.
constrain	Keeps a number inside a range: <code>constrain(7, 0, 5)</code> is 5.
% (remainder)	What's left after dividing: <code>7 % 6</code> is 1. It wraps numbers around, like a clock.

An LED strip is a screen one pixel tall.

A stadium scoreboard is rows of bulbs; a strip is one row. Snake and Pong become one-row games: left or right.

Set pixels with `incipe.setLEDcolour(...)`, push them out with `incipe.showLED()`. What `setLEDcolour` takes isn't published yet — so today the Serial Monitor is the screen.

One character per pixel, one line per frame.

On the strip	Today, on the Serial Monitor
Set one pixel's colour	Print one character: P you, * the fruit, . off
incipe.showLED() — show the frame	Serial.println() — end the line

When the LED-strip page lists its arguments, only `draw()` changes. The game logic stays exactly the same.

The joystick is the controller: reuse `zone()` and your own limits from Session 5. The code still uses the stand-in limits 25 and 75.

Each frame is a new page of a flip-book.

You don't move the drawing — you draw a new page from the facts: “you're at 0, the fruit is at 4”.

The game state is three numbers. `draw()` visits every pixel with a for loop.

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7    for (int i = 0; i < LEN; i++) {
8      if (i == player) Serial.print("P");
9      else if (i == fruit) Serial.print("*");
10     else Serial.print(".");
11   }
12   Serial.print("  score ");
13   Serial.println(score);
14 }
```

LINE 1 • STEP 1 OF 12

Our strip has 6 pixels, numbered 0 to 5.

VARIABLES

LEN = 6

SERIAL MONITOR

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7      for (int i = 0; i < LEN; i++) {
8          if (i == player) Serial.print("P");
9          else if (i == fruit) Serial.print("*");
10         else Serial.print(".");
11     }
12     Serial.print("  score ");
13     Serial.println(score);
14 }
```

LINE 2 • STEP 2 OF 12

You start on pixel 0.

VARIABLES

LEN = 6

player = 0

SERIAL MONITOR

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7    for (int i = 0; i < LEN; i++) {
8      if (i == player) Serial.print("P");
9      else if (i == fruit) Serial.print("*");
10     else Serial.print(".");
11   }
12   Serial.print("  score ");
13   Serial.println(score);
14 }
```

LINE 3 • STEP 3 OF 12

The fruit waits on pixel 4.

VARIABLES

```
LEN = 6
player = 0
fruit = 4
```

SERIAL MONITOR

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7      for (int i = 0; i < LEN; i++) {
8          if (i == player) Serial.print("P");
9          else if (i == fruit) Serial.print("*");
10         else Serial.print(".");
11     }
12     Serial.print("  score ");
13     Serial.println(score);
14 }
```

LINE 4 • STEP 4 OF 12

No points yet.

VARIABLES

LEN = 6

player = 0

fruit = 4

score = 0

SERIAL MONITOR

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7    for (int i = 0; i < LEN; i++) {
8      if (i == player) Serial.print("P");
9      else if (i == fruit) Serial.print("*");
10     else Serial.print(".");
11   }
12   Serial.print("  score ");
13   Serial.println(score);
14 }
```

LINE 7 • STEP 5 OF 12

Visit every pixel, starting at 0.

VARIABLES

LEN = 6

player = 0

fruit = 4

score = 0

SERIAL MONITOR

i = 0

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7    for (int i = 0; i < LEN; i++) {
8      if (i == player) Serial.print("P");
9      else if (i == fruit) Serial.print("*");
10     else Serial.print(".");
11   }
12   Serial.print("  score ");
13   Serial.println(score);
14 }
```

LINE 8 • STEP 6 OF 12

Pixel 0 is where you are: print P.

VARIABLES

```
LEN = 6
player = 0
fruit = 4
score = 0
```

SERIAL MONITOR

P

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7    for (int i = 0; i < LEN; i++) {
8      if (i == player) Serial.print("P");
9      else if (i == fruit) Serial.print("*");
10     else Serial.print(".");
11   }
12   Serial.print("  score ");
13   Serial.println(score);
14 }
```

LINE 7 • STEP 7 OF 12

Next pixel.

VARIABLES

LEN = 6

player = 0

fruit = 4

score = 0

SERIAL MONITOR

P

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7    for (int i = 0; i < LEN; i++) {
8      if (i == player) Serial.print("P");
9      else if (i == fruit) Serial.print("*");
10     else Serial.print(".");
11   }
12   Serial.print("  score ");
13   Serial.println(score);
14 }
```

LINE 10 • STEP 8 OF 12

Pixels 1, 2 and 3 are neither you nor the fruit: a dot each.

VARIABLES

```
LEN = 6
player = 0
fruit = 4
score = 0
```

SERIAL MONITOR

P...

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7    for (int i = 0; i < LEN; i++) {
8      if (i == player) Serial.print("P");
9      else if (i == fruit) Serial.print("*");
10     else Serial.print(".");
11   }
12   Serial.print("  score ");
13   Serial.println(score);
14 }
```

LINE 9 • STEP 9 OF 12

Pixel 4 is the fruit: print *.

VARIABLES

```
LEN = 6
player = 0
fruit = 4
score = 0
```

SERIAL MONITOR

```
P...*
```

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7      for (int i = 0; i < LEN; i++) {
8          if (i == player) Serial.print("P");
9          else if (i == fruit) Serial.print("*");
10         else Serial.print(".");
11     }
12     Serial.print("  score ");
13     Serial.println(score);
14 }
```

LINE 10 • STEP 10 OF 12

Pixel 5: a dot. i becomes 6, not below 6, so the loop ends.

VARIABLES

```
LEN = 6
player = 0
fruit = 4
score = 0
```

SERIAL MONITOR

```
P...*.
```

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;               // what you're catching
4  int score = 0;
5
6  void draw() {
7      for (int i = 0; i < LEN; i++) {
8          if (i == player) Serial.print("P");
9          else if (i == fruit) Serial.print("*");
10         else Serial.print(".");
11     }
12     Serial.print("  score ");
13     Serial.println(score);
14 }
```

LINE 12 • STEP 11 OF 12

Then the score label...

VARIABLES

```
LEN = 6
player = 0
fruit = 4
score = 0
```

SERIAL MONITOR

```
P...*.  score
```

Draw one frame.

```
1  const int LEN = 6;           // pixels on our pretend strip
2  int player = 0;              // where you are
3  int fruit = 4;                // what you're catching
4  int score = 0;
5
6  void draw() {
7      for (int i = 0; i < LEN; i++) {
8          if (i == player) Serial.print("P");
9          else if (i == fruit) Serial.print("*");
10         else Serial.print(".");
11     }
12     Serial.print("  score ");
13     Serial.println(score);
14 }
```

LINE 13 • STEP 12 OF 12

...and the score. println ends the frame.

VARIABLES

```
LEN = 6
player = 0
fruit = 4
score = 0
```

SERIAL MONITOR

```
P...*.  score 0
```

Strip Catcher: move, catch, score.

Event	What changes
Stick pushed (an edge)	player moves one pixel — but never off the strip
player lands on fruit	score goes up, fruit jumps 3 pixels on (wrapping round)
Anything changed	Draw a new frame

The curriculum task: map joystick inputs to game movements and display game state on LEDs.

The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;             // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 4 • STEP 1 OF 10

Say you're on pixel 3 and the fruit is on 4. You push right: zone 1.

VARIABLES

```
player = 3
fruit = 4
score = 0
z = 1
```

SERIAL MONITOR

The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                     // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;              // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 5 · STEP 2 OF 10

Just pushed (it was resting last pass) — go in.

VARIABLES

```
player = 3
fruit = 4
score = 0
z = 1
```

SERIAL MONITOR

The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                      // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;               // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 6 · STEP 3 OF 10

$3 + 1 = 4$, inside 0 to 5, so you move to pixel 4.

VARIABLES

```
player = 4
fruit = 4
score = 0
z = 1
```

SERIAL MONITOR



The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;             // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 7 · STEP 4 OF 10

You're on the fruit!

VARIABLES

```
player = 4
fruit = 4
score = 0
z = 1
```

SERIAL MONITOR



The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;             // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 8 · STEP 5 OF 10

score++ adds 1 to the score.

VARIABLES

```
player = 4
fruit = 4
score = 1
z = 1
```

SERIAL MONITOR

The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;              // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 9 · STEP 6 OF 10

(4 + 3) is 7, and 7 % 6 is 1: the next fruit is on pixel 1.

VARIABLES

```
player = 4
fruit = 1
score = 1
z = 1
```

SERIAL MONITOR



The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;             // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 11 · STEP 7 OF 10

Draw the new frame.

VARIABLES

```
player = 4
fruit = 1
score = 1
z = 1
```

SERIAL MONITOR

```
. * . . P .   score 1
```

The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;             // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 13 · STEP 8 OF 10

Remember: pushed. Holding the stick won't move you again.

VARIABLES

```
player = 4
fruit = 1
score = 1
z = 1
lastZone = 1
```

SERIAL MONITOR

```
.*...P.  score 1
```

The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;             // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 6 · STEP 9 OF 10

Later you're on pixel 5 and push right again: $5 + 1 = 6$ — but constrain keeps you at 5.

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VARIABLES

```
player = 5
fruit = 1
score = 1
z = 1
lastZone = 0
```

SERIAL MONITOR

```
...P.  score 1
```

The game loop, one push at a time.

```
1  int lastZone = 0;
2
3  void loop() {
4      int z = zone(incipe.getJoystickXMovement()); // from Session 5
5      if (z != 0 && lastZone == 0) {                // just pushed
6          player = constrain(player + z, 0, LEN - 1); // stay on the strip
7          if (player == fruit) {                    // caught it!
8              score++;
9              fruit = (fruit + 3) % LEN;             // next fruit, 3 pixels on
10         }
11         draw();
12     }
13     lastZone = z;
14     delay(50);
15 }
```

LINE 11 · STEP 10 OF 10

Draw: you're stuck against the end wall.

VARIABLES

```
player = 5
fruit = 1
score = 1
z = 1
lastZone = 0
```

SERIAL MONITOR

```
.*..P.  score 1
.*...P  score 1
```

A console has a restart button.

ONE LINE OF CODE

```
float pressed = incipe.getButtonResponse();
```

Button	You get
Pressed	1
Not pressed	0
Not detected	-1

Restart on release, exactly like selecting a menu item in Session 5. The IR remote joins once Session 7 can be written — its functions are not on the Wiki yet.

Make it a real game.

Step	Do this
1	Build Strip Catcher with your own joystick limits. Catch three fruits.
2	Restart: when the push button is released, reset player, fruit and score and draw a frame.
3	After 20 moves print GAME OVER and your score. Which state variable do you need to add?
4	Change LEN to 10 and the jump to $(\text{fruit} + 7) \% \text{LEN}$. Trace the first three fruit positions by hand, then check.

Checkpoint 1

LEN 6, player 0, fruit 4, score 0. What does draw() print?

READ THE CODE

```
for (int i = 0; i < LEN; i++) {  
    if (i == player) Serial.print("P");  
    else if (i == fruit) Serial.print("*");  
    else Serial.print(".");  
}
```

ANSWER

P...*. score 0

WHY

P on pixel 0, dots on 1–3, * on 4, a dot on 5, then the score.

You're on pixel 0 and push left (zone -1). Where are you now?

READ THE CODE

```
player = constrain(player + z, 0, LEN - 1);
```

ANSWER

0

WHY

$0 + (-1)$ is -1 , and `constrain(-1, 0, 5)` keeps it at 0.

You catch the fruit on pixel 4. Where is the next fruit?

READ THE CODE

```
fruit = (fruit + 3) % LEN;  
// LEN = 6
```

ANSWER

1

WHY

$(4 + 3) \% 6 = 7 \% 6 = 1.$

And if you caught it on pixel 5?

READ THE CODE

```
fruit = (fruit + 3) % LEN;  
// LEN = 6
```

ANSWER

2

WHY

$(5 + 3) \% 6 = 8 \% 6 = 2.$

Checkpoint 5

From pixel 0 you hold the stick right for 20 passes. Where are you?

READ THE CODE

```
if (z != 0 && lastZone == 0) {  
    player = ...  
}  
lastZone = z;
```

ANSWER

1

WHY

Only the first pass is an edge, so you move once.

Three things to take away.

1

An LED strip is a one-row screen: draw a frame from the game state by visiting every pixel with a for loop.

2

The joystick's edge moves you one pixel per push; constrain keeps you on the strip and % wraps the fruit around.

3

Keep game logic apart from the display: only draw() changes when the strip's calls are published.

Next — Session 10: System integration & debugging. Testing the Smart Garden and the Game Console end to end, and finding bugs with Serial prints and AI.